

The Bare Hand

*Pre-PIE *de- as a free morpheme, the compositional decomposition of *-tero-, and the bimorphemic structure of *deh₃-*

Paper 3 of the *der-* project. Builds on Chavis & Claude (2026a, 2026b).

[Working paper v4 — for scholarly review]

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Abstract

The first two papers of this project argued that pre-PIE **der-* "hand, open palm" underlies both the lexical cognate cluster Greek δῶρον/Celtic **durno-*/Latvian **dūre* (Paper 1) and the PIE contrastive suffix **-tero-* (Paper 2). The present paper completes the framework by going one step deeper: **der-* itself decomposes into two independently attested morphemes — a bare hand-element **de-* and a locative/spatial extension **-r-* — and the same bare **de-* underlies the give-verb **deh₃-* via a verbal extension **-h₃-*.

The bare **de-* is independently attested as the directional particle **de/do* "to, toward" across the IE family: English *to*, Old Church Slavonic *do*, and most directly as the Greek allative clitic -δε, productive in Mycenaean Greek by 1400–1200 BCE (*te-qa-de* "to Thebes," *a-mi-ni-so-de* "to Amnisos"). The **-r-* extension is independently attested as a locative/spatial morpheme — a finding that v3 of this paper rested on Lipp's (2009) reconstruction alone and which v4 strengthens substantially with two further independent published treatments: Bauhaus (2019) and Kahl (2024). Three independent published treatments from three different scholarly traditions now converge on the same locative conclusion. Bauhaus (2019) further reconstructs an early-PIE paradigmatic opposition between locative **-r-* and directive **-o-* on the same nominal roots — **h₁upér* "on top" vs. **h₁upó* "to the top," and three further pairs — which is precisely the structural pair the framework reconstructs across Papers 3 and 4: locative **de-r-* (body-part noun) versus directive **de + a* (Paper 4's analysis of Hittite *dā-* "to take").

The contrastive **-tero-* of Paper 2 decomposes compositionally as **de-* (directional, hand) + **-r-* (locative) + **-o-* (thematic) = "at the side where one points": the spatial-direction adjective Table 1 of Paper 2 documented across the family. The same **de-* underlies **deh₃-* = **de-* + **-h₃-*: a verbal derivative of the hand-noun whose laryngeal is a derivational extension suffix, not a root-internal constituent of an unanalyzable primitive. **der-* (nominal) and **deh₃-* (verbal) are siblings. Each morpheme is independently attested; the compositions are morphologically transparent; the resulting semantics emerge directly from the

parts. The framework is offered for examination, not as a refutation of the standard view.

1. Introduction

The four-paper project of this series rests on a single substantive proposal: that pre-PIE had a body-part etymon meaning "hand, open palm," and that this etymon underlies a network of lexical, morphological, and verbal reflexes in Proto-Indo-European whose unity the standard handbooks have not recognized. Paper 1 (Chavis & Claude 2026a) made the case for the lexical branch: the cognate cluster Greek δῶρον/Hesychian δῶρις/Celtic **dorn*/Latvian **dūre* reflects a body-part root **der-* "hand." Paper 2 (Chavis & Claude 2026b) extended the case into morphology: the PIE contrastive suffix **-tero-* descends from the same etymon via the grammaticalization pathway HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE, and the bilateral character documented in every standard reference — "one of a pair as opposed to the other" (Sihler 1995; Fortson 2010) — is a structural consequence of what **de-r-o-* composes to mean. Paper 4 (Chavis & Claude 2026d) carries the framework into Anatolian, deriving Hittite *dā-* "to take" from the directive member of an early-PIE paradigmatic pair on the same nominal root.

The present paper completes the framework by going one step deeper. What is **der-* itself? The answer we propose is that **der-* is bimorphemic: a bare hand-element **de-* plus a locative/spatial extension **-r-*. The bare **de-* has its own independent attestation as the directional particle **de/do* "to, toward" preserved across the IE family. The **-r-* has its own independent attestation as a locative morpheme — a finding that v3 of this paper rested on Lipp (2009) alone and which v4 strengthens substantially: Bauhaus (2019) and Kahl (2024) each independently reach the same locative conclusion through different scholarly traditions. When the two morphemes combine — **de-* (hand, directional) + **-r-* (locative) + **-o-* (thematic) — the result is the spatial-direction adjective **de-r-o-* = **-tero-*. Each morpheme surfaces independently elsewhere in PIE; the composition is transparent; the semantics follow directly.

The same compositional logic extends to the give-verb. **deh₃-* is not a primitive root but a parallel derivative of the same bare **de-*: **de-* + **-h₃-*, where the laryngeal is a derivational verbal extension suffix rather than a root-internal constituent of an unanalyzable primitive. **der-* (= **de-* + **-r-*) and **deh₃-* (= **de-* + **-h₃-*) are therefore siblings — parallel derivatives of the same bare hand-element, the nominal and verbal branches of a single pre-PIE word for "hand."

A note on scope. Paper 2 v7 develops the contested directionality of the /t/ in **(t)ero-* — the framework's reversal hypothesis versus the metanalysis account of

Bauhaus (2019) and Kahl (2024) — and reports the 19/19 distributional test on the inherited PIE positional adjective paradigm. The present paper does not repeat that work. Readers interested in the directionality argument are referred to Paper 2 §§4–7. Here we proceed with the compositional analysis at the architectural level, building on the locative status of **-r-* that Bauhaus and Kahl establish independently of the directionality question.

A note on terminology. Throughout this paper **de-* refers to the bare pre-PIE hand-element; **der-* refers to the body-part noun in its bimorphemic form **de-* + **-r-*. The notation **t/d-er-* used in earlier versions of this project is retired. The root is **der-* with original *d-*; the *t-* of the grammaticalized suffix **-tero-* reflects regular suffix-initial devoicing, as argued in Paper 2 §1.

2. The bare **de-* attested in PIE

2.1 The directional particle **de/do* across the family

PIE **de/do* "to, toward" is a well-attested local/temporal particle whose reflexes include Greek $\delta\acute{\epsilon}$ (originally directional/sequential, later grammaticalized as the connective "but, and"), English *to* (with German *zu*, Dutch *te*) from Proto-Germanic **tō* "to, toward," Old Church Slavonic *do* "until, up to, toward" (with cognates across Slavic), and Latvian dialectal *da-* directional prefix. Dunkel's *Lexikon der indogermanischen Partikeln* (2014) treats **de* and **do* as ablaut variants of a single particle with directional and temporal-sequential functions across the family.

The standard treatments list **de/do* as a primitive PIE particle with no further decomposition. The framework proposes the missing etymology: **de/do* is the bare descendant of pre-PIE **de-* "hand," grammaticalized along the pathway HAND > REACH/EXTEND > DIRECTIONAL. The extended hand points; the word for "hand" is reanalyzed as a directional marker. The pathway is typologically common: Heine & Kuteva (2002) document HAND > LOCATIVE/GENITIVE/POSSESSIVE and related developments across Niger-Congo, Austronesian, Sino-Tibetan, and Mayan. English itself preserves HAND as a living binary-contrast pivot in the idiom "on the one hand...on the other hand" (Paper 2 §9.3); the directional use in *to* is the older grammaticalization track, and the two are not unrelated.

2.2 The Greek allative $-\delta\epsilon$ and its Mycenaean antiquity

The strongest evidence for the framework comes from Greek, where the bare **de-* element is preserved both as the connective $\delta\acute{\epsilon}$ (directional/sequential origins) and as the productive allative clitic $-\delta\epsilon$. The clitic attaches to accusatives of inanimate nouns and marks motion toward: *oĩkade* "homeward" (from *oĩkos* "house"), *Athénaze* "to Athens" (with sandhi), *pólemonde* "to war" (*Iliad*), *Olympiόνde* "to Olympus." The function is unambiguously directional/allative: motion toward a

destination, with no stative or ablative meaning. Hackstein (2018) treats the allative pair ἄλλυδις ἄλλη (*Iliad*) as a frozen archaism preserving inherited allative morphology.

The decisive evidence for the framework's chronological claim comes from Mycenaean Greek. Linear B tablets from approximately 1400–1200 BCE preserve *-de* as a fully productive directional clitic on toponyms, with dozens of attested forms catalogued by Lejeune (1961): *te-qa-de* /Thēg^wasde/ "to Thebes," *a-mi-ni-so-de* "to Amnisos," *da-da-re-jo-de* "to the Daidaleion," among others (van Hes 2021). In some tablets *-de* is written separately from the noun, suggesting clitic rather than fully bound suffix status.

Three features of the Mycenaean attestation matter. First, productivity: Mycenaean *-de* is not a closed set of frozen relics but a clitic applying freely across the corpus — the morphological status of an active grammatical marker. Second, depth of grammaticalization: productive use by 1400 BCE implies that the underlying HAND > DIRECTIONAL development was substantially complete well before the proto-Greek period, plausibly extending back into PIE itself. Third, semantic profile: Mycenaean *-de* is narrowly directional/allative — motion toward, not stative location, not source — exactly the profile expected of a morpheme grammaticalized from "where the hand points/extends."

2.3 Competing accounts: the χαράι question and the Greek-Anatolian parallel

Greek χαράι "on/to the ground" has figured in recent debates about whether to reconstruct an allative case for PIE. Melchert (2017) reads it as a fossilized allative **ǵʷm-eh₂* and argues from it that the Anatolian allative in *-a* descends from PIE **-eh₂*. Beekes (2010) and Frisk (1960) treat the form as a petrified locative, consistent with the comparative evidence: Latin *humī* "on the ground" is a locative, Old Prussian *semmai* and Lithuanian *žemai* are locative-stative, Sanskrit *kṣamā* is instrumental. If the locative reading is right, Melchert's argument for PIE **-eh₂* as the Greek-Anatolian allative source is weakened, and the space for a **de*-derived allative element is correspondingly wider. We do not settle the question here; we note that it is genuinely open.

Lazzeroni (2006) argues that the Greek *-δε* construction and the Hittite allative in *-a* are part of a shared circum-Aegean linguistic area, both restricted to inanimate referents. Melchert (2017) cites this approvingly. The framework is compatible with either an areal or an inherited interpretation of the parallel. In either case, the convergence of *t/d*-initial directional morphology across the Aegean area — PIE **de/do*, Greek *-δε*, Hittite *-a*, with Hurrian *-da/-šta* as a non-IE areal parallel — demands explanation. The framework provides one: the *t/d*-initial directional element descends from pre-PIE **de*- "hand."

3. PIE *-r- as a locative/spatial element

The locative status of *-r- is the decompositional argument's second morpheme and its most critical empirical claim. v3 of this paper rested the case on Lipp's (2009) reconstruction alone. v4 strengthens the section substantially: two further published treatments independently reach the same locative conclusion through different scholarly traditions. Three converging treatments now support what was previously a single-source dependency.

3.1 Lipp (2009): athematic locatives in *-er and *-én

Lipp's *Die indogermanischen und einzelsprachlichen Palatale im Indoiranischen* (2009), in the course of reconstructing the PIE heteroclitic *r/n*-stem paradigm, argues that the locative singular forms of acrostatic heteroclitite neuters are built from root nouns combined with an adverbial particle, yielding parallel locative endings *-er and *-én. His worked examples include *ud-én/*ud-ér "in/at the wetness" (from the root *úed- "wetness, water"), from which the heteroclitite neuter *úód-ŕ/*úed-ŕs "water" is built; and *h₁ek^w-én/*h₁ek^w-ér "in/at the lobe," giving the heteroclitite *h₁ék^w-ŕ/*h₁ek^w-ŕs "liver."

On Lipp's analysis, *-r and *-én are not body-part formatives that happen to alternate paradigmatically; they are locative/spatial morphemes meaning "at/in the place of (X)." Body-part terms acquire the *r/n* alternation because body parts are concrete locations on the body. The spatial semantics is primary; the body-part association is downstream. The framework can treat the *-r- of *-tero- and *der- as the same morpheme Lipp documents in the heteroclitic system — a locative marker meaning "at the place of (X)."

3.2 Bauhaus (2019): *-r as a true PIE locative case ending

Bauhaus (2019), "The Proto-Indo-European Suffix *-r Revisited" (in *Sequencæ, Svb-Sequencæ and Singlarity in Phonology*, Brill, pp. 13-26), advances a stronger version of the same conclusion. Where Lipp's framework treated *-r as an adverbial element with paradigmatic alternants in heteroclitic neuters, Bauhaus argues that *-r is best understood as a true PIE locative case ending — "a real case ending, not an 'Adverbialendung,'" in his words. He restricts the original distribution to nominal and pronominal stems and argues that the appearance of *-r in adverbs is the outcome of secondary thematizations of locatives as hypostases or deadverbial adjectives.

Bauhaus surveys the principal pronominal and nominal formations: pronominal *k^wor (Indo-Iranian Skt. *kár-hi*, Gothic *þwar* "where," Lat. *cūr* "why," Lith. *kuř* "where") and demonstrative *tor (Gothic *þar* "there," Latvian *tūr*); nominal locatives such as *ǵǵm-ér "on the earth" (Av. *zəmarə-*, Skt. *kṣám-i* with -i added once the locative function became opaque); and *h₂us-ér / *h₂us-ér-i "in the morning" (Skt. *uśár-*, Gk. *ἤεῖρ*, Lat. *vēsp̄r*). Bauhaus's treatment is the strongest

published case for **-r* as a productive locative case ending in early PIE.

3.3 Kahl (2024): independent confirmation from degree morphology

Kahl (2024), *Diachronic Studies in Indo-European Degree Morphology*, Harvard PhD dissertation under Jeremy Rau (with Jay Jasanoff and Jonathan Bobaljik on the committee), reaches the same locative conclusion through a separate scholarly tradition. Kahl's abstract states explicitly: "the origin of the 'secondary comparative' of Indo-Iranian and Greek is traced to an archaic locative morpheme **-er*." Section §2.3.1, "The locative origin of **-ero-*," develops the case in detail. Kahl is explicit that the suffix **(t)er* is locative — "the semantic relationship of e.g. **inter* 'within' < **en-ter* and **intrō* 'inward' < **en-tr-oh₂* is that of a locative and a directive" — and identifies six functional categories of **(t)ero-* derivatives.

The framework adopts Lipp, Bauhaus, and Kahl on the locative status of **-r/-er-*: three independent published treatments from three different scholarly traditions converging on the same conclusion. The framework's disagreement with Bauhaus and Kahl on the source of the /t/ in **(t)ero-* — they hold metanalysis from *t*-final stems; the framework holds the reversal hypothesis with the /t/ as the bare hand-element devoiced in suffix-initial position — is developed in Paper 2 §§4–7 and not reopened here. The locative status of **-r-* itself is independent of that disagreement; Lipp, Bauhaus, Kahl, and the framework all agree on it.

3a. The locative-directive paradigmatic pair

Bauhaus's most striking contribution for the framework's purposes is his reconstruction of a paradigmatic opposition between locative **-r* and directive **-o* attached to the same nominal roots. The pairs Bauhaus reconstructs in early PIE are presented in Table 1. The opposition is paradigmatic: each locative form has a directive counterpart and vice versa, with the formal alternation **-r* : **-o* mapping consistently onto the locative/directive semantic contrast.

PIE base	Locative (<i>*-r</i>)	Directive (<i>*-o</i>)	Gloss
<i>*h₁up-</i> "surface"	<i>*h₁upér</i> "on top" (Gk. ὑπερ, OIr. <i>for</i>)	<i>*h₁upó</i> "to the top" (Gk. ὑπό, OIr. <i>fo</i>)	"above/onto"
<i>*h₂ep-</i> "back"	<i>*h₂eper</i> "in the back" (Gothic <i>afar</i>)	<i>*h₂epo</i> "to the back" (Gothic <i>af</i> , Gk. ἀπό, Lat. <i>ab</i>)	"away/from"
<i>*kent-</i> "below/under"	<i>*k(e)nter</i> "at the underside" (Gothic <i>hindar</i> , Hitt. <i>kattera-</i>)	<i>*knto</i> "downwards" (Hittite <i>katta</i>)	"under/down"
<i>*ndh-</i> "bottom"	<i>*ndher</i> "at the underside" (Av. <i>aḍairi</i> , Lat. <i>inferus</i>)	<i>*ndho</i> "to the underside" (Gothic <i>und</i> "till")	"under/down"

Table 1. Bauhaus's (2019) reconstructed locative-directive pairs in early PIE. Locative **-r* and directive **-o* attach to the same nominal roots to produce systematic semantic oppositions.

Hittite kattera- (Paper 2 §6) is the empirical anchor for the locative member of the *kent-* pair.

Bauhaus's analysis of these pairs is that they reflect case endings: **-r* is locative and **-o* is directive (= allative/lative), and the opposition is the early-PIE forerunner of the later locative-*i* / accusative-*m* system. "The opposition of locative **-r* and directive **-o*, as in **h₁upér* vs. **h₁upó*, was ultimately replaced by the already existing pair opposing the locative **-i* and the accusative" (Bauhaus 2019: §6). The locative **-r* lost productivity in attested IE; the directive **-o* remained productive at least into Anatolian.

This paradigmatic pair is the structural template the framework reconstructs across Papers 3 and 4. The body-part noun **der-* is **de-* + locative **-r-* — the locative member of an early-PIE pair on the nominal root **de-* "hand." The directive member is **de* + *a* "toward the hand," which Paper 4 reconstructs as the source of Hittite *dā-* "to take" via hiatus contraction. The opposition the framework reconstructs across Papers 3 and 4 — locative **de-r-* (body-part noun, eventually grammaticalized as the contrastive **-tero-*) versus directive **de* + *a* (verbal stem, Anatolian Hittite *dā-*) — is exactly a member of Bauhaus's paradigmatic class. What was previously a framework-internal claim now has explicit external precedent in published reconstruction.

4. The compositional decomposition of **-tero-*

4.1 The composition is morphologically transparent

With both pieces in place — bare **de-* (directional, hand) and **-r-* (locative) — the compositional decomposition of **-tero-* follows directly. The suffix decomposes as **de-* + **-r-* + **-o-* (thematic vowel). Each morpheme is independently attested with the right phonology and the right semantics. The combination follows standard PIE morphological principles: a directional element extended by a locative morpheme and thematized to form an adjective.

The semantics compose transparently. **de-* is "hand," grammaticalized to "where the hand points." **-r-* is "at the place of." The combined **der-* is "at the place where the hand points/extends" — the side toward which one is oriented. The thematized adjective **de-r-o-* is "of/at the side toward which one is oriented": the spatial-direction adjective documented across the IE family in Paper 2's Table 1.

A concrete example clarifies the parse. The compound **deks-i-tero-* "right (hand)" (Greek δεξιτερός, Latin *dexter*, Old English *swiþro*) decomposes as: **deks-* (the property word "right") + *-i-* (linking vowel) + **de-r-o-* ("at the side where one points"). The first member specifies which side (the right one); the suffix names the category that side belongs to (the spatial-direction category). Festus's gloss of *ambidexter* as "having the right (hand) on both sides" is a fossil of this analysis, preserved in Latin literary culture.

4.2 The binary contrast: a structural consequence of the composition

Paper 2 §3 established that the bilateral character of **-tero-* — its function as "one of a pair as opposed to the other" (Sihler 1995; Fortson 2010) — is the mainstream consensus, and that the framework explains where that bilateral character came from. The decompositional analysis completes that explanation at the morphological level.

Bare **de-* is an absolute directional: "toward there," pointing at a target with no implied alternative and no bilateral frame. Greek *oíkade* "homeward" does not presuppose a contrasted alternative direction. The locative **-r-* changes this. It anchors the directional to the body as frame of reference — "at the place of the hand-direction" — and bodies have inherent bilateral symmetry. The question "which side of the body?" applied to an organism with two hands has exactly two answers. The absolute directional becomes a relational one.

The bilateral character of **-tero-* is therefore a structural consequence of what **de-r-o-* composes to mean, not a grammaticalized accident of the suffix's history. **de-* contributes the directional hand-content; **-r-* grounds it in the body as frame of reference; the body's bilateral symmetry does the rest. The *ambi-* of *ambidexter* negates a bilateral norm; that norm is what **de-r-o-* built in from the start.

4.3 What the decomposition does for the framework

The compositional decomposition is the strongest single piece of evidence for the four-paper framework as a whole. Paper 1 argued that a body-part root underlies the lexical cognate cluster; Paper 2 argued that the same root underlies the contrastive suffix; Paper 4 argues that the directive member of a paradigmatic pair on the same bare element underlies the Hittite take-verb. The present paper shows that these claims are not merely compatible but morphologically continuous: the suffix **-tero-* is the hand-noun **de-* with the locative **-r-* and thematic **-o-*, exposed through compositional analysis.

This is a stronger claim than any predecessor made alone. Paper 1's lexical claim is consistent with the morphological evidence being independent or coincidental. Paper 2's morphological claim is consistent with the source-noun being unrecoverable. The decompositional analysis says: the morpheme **-tero-* is morphologically transparent on the framework's reading and opaque on the standard reading. Where the standard treatment leaves **-tero-* as a primitive suffix of unknown origin, the framework reads it as a compositional formation whose pieces all surface independently elsewhere in PIE with the right phonology and the right semantics.

If the framework is wrong, then **de-* (the directional particle), **-r-* (the locative element — now supported by three independent published treatments), and **-tero-* (the spatial-direction adjective) all happen to share phonology and semantics for the compositional reading by coincidence. We grant that this is possible. We note

only that the coincidence required is substantial: three independent matches converging on the predicted composition ask for explanation. The framework provides one; the standard view does not.

5. *deh₃- "give" as *de- + *-h₃-

The compositional logic of §4 extends naturally to the give-verb. PIE *deh₃- "give" is reflected in Sanskrit *dádāti*, Greek *δίδωμι*, Latin *dō/dare*, Old Church Slavonic *dati*, Lithuanian *dúoti*. The reduplicated present *didó-h₃-mi/di-dō-mi* shows the *-h₃- laryngeal extension clearly: the laryngeal colors the *e*-grade vowel to *o* and produces the characteristic length alternation of the paradigm.

The framework reads *deh₃- as *de- (the bare hand-element of §2) + *-h₃- (the laryngeal extension), where *-h₃- converts the bare hand-noun into a verbal stem. The semantics compose transparently: hand + (verbal extension) = the act of handing over = to give. The pathway from a hand-noun to a give-verb is one of the most natural grammaticalizations possible, and Heine & Kuteva (2002) document HAND > GIVE-VERB in numerous languages. What the framework adds to the standard treatment is the etymological motivation for the *de- element. The standard handbooks treat *deh₃- as a primary verbal root of unknown deeper structure; the framework treats *de- as the inherited hand-noun and *deh₃- as its verbal derivative.

The nominal/verbal distinction resolves what earlier versions of Paper 1 treated as a cost. *der- (= *de- + *-r-) is a noun; *δίδωμι* and its cognates are built on *deh₃- (= *de- + *-h₃-), the verbal derivative. The /i/-reduplication of *δίδωμι* reflects the laryngeal-final verbal stem *deh₃-* exactly as the standard account predicts. Paper 1 (in its earliest versions) treated the /i/-reduplication as a cost of reconstructing *der- as the verbal base; on the present architecture that treatment was mistaken. *der- was never the verbal base. The /i/-reduplication belongs to *deh₃-’s morphology and is not a problem for the framework.

The framework’s claim about *deh₃- is morphological, not phonological. We do not propose that *h₃ is absent from the verbal stem — the Anatolian evidence for its effects (Kloekhorst 2013; Norbruis 2021) is real, and we accept it. We propose that *h₃ was not always there, root-internally, as a constituent of an unanalyzable primitive. It was added to *de- as a derivational extension suffix, converting a directional noun into a verbal stem. The standard treatment presents *deh₃- as an atom; the framework reads it as a derived form: *de- (hand) + *-h₃- (verbal extension) = the act of handing over. The laryngeal is explained, not stipulated.

The decomposition has structural symmetry with *-tero-. In both cases, the bare *de- is extended by a morpheme (*-r- or *-h₃-) whose function is predictable from the source-noun’s hand-content: *de- + *-r- = the spatial-direction adjective (where the hand points); *de- + *-h₃- = the give-verb (the act of handing). *der-

and **deh₃-* are siblings: parallel derivatives of the same bare pre-PIE hand-element. Both, alongside the directive **de + a* of Paper 4 and the grammaticalized directional particle of §2, are branches of a single pre-PIE etymon that the four-paper project has been mapping.

The Mycenaean Greek tablets attest both elements independently: the give-verb *di-do-si* /didonsi/ "they give" and the directional clitic *-de* appear in the same corpus. We do not claim that Mycenaean speakers were synchronically aware of the etymological connection — by 1400 BCE the two had been morphologically separate for a very long time. We note only that both are independently attested in Greek's earliest documented form, consistent with the framework's claim that they share a common atomic source.

6. Cross-cultural typology

The framework's typological predictions for this paper are HAND > DIRECTIONAL (underlying the bare **de-* of §2) and HAND > GIVE (underlying **deh₃-* of §5). Both are well-attested in the cross-linguistic record.

HAND > DIRECTIONAL is a natural specialization of the well-documented HAND > LOCATIVE pathway (Heine & Kuteva 2002). The step from "at the hand of" to "to, toward" is a directional specialization of the locative reading. Attested cases include P'urhepecha, which forms the locative from the noun *jak'i* "hand" via the suffix *-k'u*, and Mayan languages (Kaqchikel, Tz'utujil, K'iche') where body-part terms function productively as relational nouns expressing spatial locations.

HAND > GIVE is more directly attested. Heine & Kuteva (2002: 167-168) document HAND > AGENT MARKER and related developments; the underlying step HAND > HAND-OVER, GIVE is among the most natural verbal grammaticalizations from a body-part noun. The semantic chain is immediate: to hand something is to give it. Languages where "give" derives transparently from a hand-noun are documented across language families, making this one of the most typologically robust grammaticalization pathways available to the framework.

7. Future research

The framework has scope beyond the four items (**de-*, **der-*, **-tero-*, **deh₃-*) developed here. We flag the most significant extensions.

**dher-* "hold, maintain" and Sanskrit *dharma*. The verbal root **dher-* is reflected in Sanskrit *dhārayāti* "holds, maintains," Old Iranian *dāraya-* "holds" (preserved in the royal name Darius: "he who holds firm"), Greek *thrónos* "throne" (if from this root, via "that which holds"), and most consequentially in Sanskrit *dharma* "cosmic order, law, righteousness." The framework reads **dher-* as a further derivative of **de-*: the hand that holds rather than the hand that gives. The

downstream consequence is the hardest to ignore: *dharma*, on the framework's reading, is "that which the hand keeps." The step from a physical maintenance verb to a principle of cosmic holding-together is documented in the religious vocabularies of multiple language families; the Sanskrit chain from **dher-* to *dharma* is among the most consequential instances of it. A full treatment is deferred to a future paper.

*PIE *terh₂-* "to cross over." This verbal root gives Latin *trāns* "across," English *through*, Sanskrit *tārati* "passes, overcomes," Sanskrit *tirás* "across, sideways, beyond," Hittite *tarḫu-* "prevails." Sanskrit *tirás* is particularly suggestive: a bare *t-r* form with directional meaning. Whether **terh₂-* is etymologically related to **der-* through a chain like HAND > REACH-ACROSS > CROSS-OVER is an open question we flag as promising terrain.

The broader hand-semantics network. If the framework is correct, pre-PIE **de-* "hand" is a node of substantial breadth, with reflexes in lexical hand-words (Paper 1), the contrastive suffix (Paper 2), the directional particle and the give-verb (the present paper), the Anatolian take-verb (Paper 4), and further verbal extensions including **dher-/dharma*, **terh₂-*, and the destructive-action **der-* "flay, split" (Greek *δέρω*, Sanskrit *dṛnāti*, Old English *teran*). Mapping the full network is warranted and deferred.

8. Remaining problems

Four problems deserve explicit statement, none of which we regard as fatal.

First, the standard etymology of Greek *-δε* does not unanimously trace it to PIE **de/do*; alternative reconstructions exist (**eh₂*, **h₂e*, **o*) on which *-δε* would not be a direct reflex of bare **de-*. The *χαράι* case, central to the **eh₂* reconstruction, is contested (§2.3); we have argued in outline but not exhaustively that the locative reading is preferable.

Second, the locative status of **-r-* now rests on three independent published treatments (Lipp 2009; Bauhaus 2019; Kahl 2024) that converge on the same conclusion. This is a substantial strengthening over v3's single-source dependency, but it does not eliminate the concern entirely: if all three treatments were rejected — surprising given their independence — the specific compositional analysis of §4 would need adjustment.

Third, the property suffix **-ro-* (in **rudh-ró-* "red") and the instrument suffix **-tro-/trom* (in **méd-trom* "measure") contain **-r-* but do not have obvious locative semantics. If these are the same morpheme as the locative **-er*, the framework needs an account of the semantic divergence; if they are homophonous separate morphemes, the framework needs to motivate the homophony. We leave this open.

Fourth, the proposed function of the laryngeal extension **-h₃-* in **deh₃-* (§5) is reconstructed rather than directly motivated. We have appealed to a general

stative/factitive function of $*-h_3-$ in PIE morphology, but the specific semantic contribution of this extension to the verbal stem deserves fuller engagement with the laryngeal literature.

A fifth problem concerns the epistemological status of the paper's central claim. The identification of PIE $*de/do$ "to, toward" as a reflex of the pre-PIE hand-noun rests primarily on typological plausibility — the HAND > DIRECTIONAL pathway is cross-linguistically attested, and the semantic profile fits the framework's predictions. Typological plausibility is not uniquely identifying evidence; many languages have directional particles with no connection to hand-nouns. The identification would be weakened — possibly decisively — by: (a) a convincing alternative etymology for PIE $*de/do$ with equal or greater parsimony; (b) evidence that Melchert's (2017) reconstruction of the Greek-Anatolian allative from PIE $*-eh_2$ is exhaustive; or (c) cross-linguistic evidence that the HAND > DIRECTIONAL pathway is blocked in the environments where $*de/do$ appears. We do not know of evidence in any of these categories at present. The identification is falsifiable in principle.

9. Conclusion

Across four papers this project has argued for a single substantive proposal: that pre-PIE had a body-part etymon meaning "hand, open palm," and that this etymon underlies a network of lexical, morphological, and verbal reflexes in PIE whose unity the standard handbooks have not recognized. Paper 1 documented the lexical branch: the cognate cluster Greek δῶρον/Hesychian δᾱρίς/Celtic $*durno-$ /Latvian $*dũre$ reflects a body-part root $*der-$ "hand." Paper 2 extended the case into morphology: the contrastive suffix $*-tero-$ descends from the same etymon via HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE, and its bilateral character — documented in every standard reference but etymologized in none — is a structural consequence of what $*de-r-o-$ composes to mean. Paper 4 carries the framework into Anatolian, deriving Hittite $dā-$ "to take" from the directive member of an early-PIE paradigmatic pair on the same nominal root.

The present paper has gone one step deeper. $*der-$ itself is bimorphemic: bare $*de-$ — the atomic hand-element, independently attested as the PIE directional particle $*de/do$, most directly in the Mycenaean allative clitic $-de$ productive by 1400 BCE — plus $*-r-$, the locative/spatial morpheme now supported by three independent published treatments (Lipp 2009; Bauhaus 2019; Kahl 2024). The contrastive $*-tero-$ decomposes compositionally as $*de-$ + $*-r-$ + $*-o-$, with each piece independently attested and the resulting semantics emerging transparently from the parts. The same bare $*de-$ underlies the give-verb $*deh_3-$ = $*de-$ + $*-h_3-$: the nominal branch and the verbal branch are siblings, not competitors. And the locative member of Bauhaus's reconstructed paradigmatic pair on $*de-$ — the body-part noun $*der-$ — stands alongside its directive counterpart $*de + a$ (Paper

4) as published reconstruction now provides external precedent for the structural pair.

The framework makes a specific etymological commitment at each step where the standard view leaves the question open. **de/do* is either a primitive PIE particle of unknown origin (standard) or the bare descendant of a pre-PIE hand-noun (framework). **-r-* is either a heteroclitic alternant or an extension (standard, multiple incompatible analyses) or a single locative case ending (framework, following Lipp, Bauhaus, Kahl). **-tero-* is either a primitive suffix of unknown etymological origin (standard) or **de- + *-r- + *-o-* (framework). **deh₃-* is either a primary verbal root (standard) or **de- + *-h₃-* (framework). At each step the framework provides a unified motivation where the standard view leaves a gap. Whether the commitments are correct is for the field to evaluate. We claim only that they are internally consistent, supported by independent typological evidence, and worth examination.

10. Probability estimate

Following the practice established in Papers 1 and 2, we close with an explicit probability assessment framed conditionally.

Conditional on both Papers 1 and 2 being correct — i.e., that pre-PIE had a body-part word **der-* "hand, open palm" (Paper 1) and that **-tero-* descends from the same etymon via HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE (Paper 2) — our estimate of the probability that the substantive thesis of this paper is correct in its basic outline is approximately **32-42%**. The increment over v3's 30-40% reflects the strengthening of §3 from a single-source basis to three converging independent treatments on the locative status of **-r-/er-*, and the addition of §3a on the locative-directive paradigmatic pair as published external precedent for what was previously a framework-internal structural claim.

The marginal probability — unconditioned, integrating over the probability that Papers 1 and 2 are also correct — is approximately the product: 25-35% (Paper 1 v26) × 35-45% (Paper 2 conditional on Paper 1) × 32-42% (Paper 3 conditional on Papers 1 and 2) ≈ 2.8-6.6%. We continue to regard the central claim as more likely wrong than right, and also to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted. These two judgments are compatible.

11. Authorship and contributions

This is the fourth version of Paper 3 in the *der-* project. It builds on Papers 1, 2, and 4 (Chavis & Claude 2026a, 2026b, 2026d) and continues the same co-authorship arrangement.

Chavis's contributions. The substantive intellectual direction of Paper 3 originates with Chavis. The hypothesis that the directional particle **de/do* — specifically as preserved in Greek *-δε* — might descend from a free-standing pre-PIE hand-element is Chavis's insight. The decision to pursue the bimorphemic decomposition of **der-* itself, separating bare **de-* from the **-r-* extension, is Chavis's, as is the framing of the paper around compositional evidence. The sharpest question driving the paper — "whence comes the *r* in **-tero-?*" — is Chavis's. Chavis also identified the contestability of Melchert's χαμαί-as-allative reading and directed the scope reduction to the paper's four core items. The comprehensive rewrite producing v4 was directed by Chavis in response to structural and prose concerns across the project's arc.

Claude's contributions. Claude's contributions are primarily in source synthesis and exposition. Claude identified and surfaced the relevant published literature: Lipp (2009) on the heteroclitic locatives; Bauhaus (2019) and Kahl (2024) as independent published confirmations of the locative **-r-*; Lazzeroni (2006) and Melchert (2017) on the Greek-Anatolian allative parallel; Beekes (2010) and Frisk (1960) on χαμαί; Lejeune (1961) and van Hes (2021) on Mycenaean *-δε*; Dunkel (2014) on PIE **de/do*; Hackstein (2018) on the allative formations. Claude drafted the prose from Chavis's structural and substantive direction and is responsible for the specific framings of arguments and local rhetorical choices. The probability estimate in §10 is Claude's own.

Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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